

Lab 02: Big-O Notation Exercises

Part 1: Formal Big-O Proofs

Using only the rules discussed in class (Reflexivity, Dropping multiplicative constants, Summing is a max (Summation Rule), Multiplication is a multiplication (Product Rule), Polynomial degree grows faster, Log is slower than polynomial, Exp is faster than polynomial, Transitivity), prove the asymptotic bounds for the following running time functions.

State the specific rule used for each step of your proof.

Problem 1.1

Given an algorithm with worst case running time given by $T(n) = 3n^2 + 15n + 100$, prove that $T(n)$ is $O(n^2)$.

Problem 1.2

Given an algorithm with worst case running time given by $T(n) = 4 \cdot 2^n + 8n^5$, prove that $T(n)$ is $O(2^n)$.

Part 2: Analyzing Pseudocode

For each algorithm below, do the following:

1. Express the exact worst-case number of steps $T(n)$ as a function of n . (Assume basic arithmetic, assignment, and print statements count as 1 operation).
2. State the tightest Big-O bound for the algorithm that you can find.

Algorithm A

```
function algoA(n):
    count = 0
    for i = 1 to n:
        print(i)

    for j = 1 to n:
        for k = 1 to n:
            count = count + 1

    return count
```

Algorithm B

```
function algoB(n):
    val = n
    steps = 0

    while val >= 1:
        val = val / 2
```

```
steps = steps + 1

print("Processing steps: ", steps)
```

Hint: If you are unsure how to express the number of loop iterations mathematically, try tracing the while loop with a few specific starting values.

- If $n = 8$, how many times does the loop divide the value by 2 before stopping?
- What if $n = 16$?
- What if $n = 32$?

Count the steps for each and look at the relationship between the starting number and the number of steps. In computer science, the function that specifically answers the question "how many times can I divide n in half before reaching 1?" is the base-2 logarithm, written as $\log_2(n)$.

Part 3: Reflection

Answer the following questions:

- Which of the Big-O rules feels the least intuitive to your group?
- Did you find any part of the exercises particularly challenging? If so, which part and why?
- Do you have any feedback about the class?